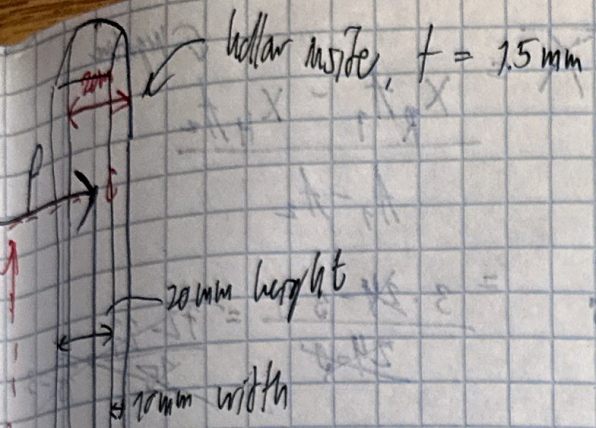
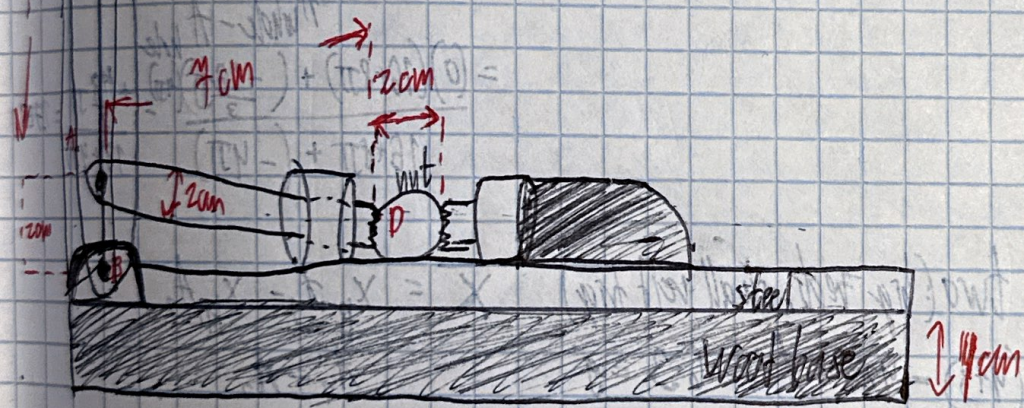




Given: macadamia nut: 13-20 ϕ
 average grip strength $\approx 2300 \text{ N}$
 * it takes $220 \text{ kg} \approx 2200 \text{ N}$
 to crack a macadamia nut



FBDs: Will this crack the nut?

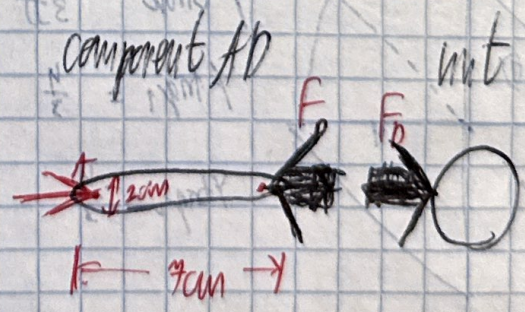
"explode" the system:

component BAC

$$\sum M_B = 0 = 300 \text{ N} \cdot 0.22 \text{ m} + 0.02 \text{ m} \cdot F_A$$

$$F_A = -\frac{300 \text{ N} \cdot 0.22 \text{ m}}{0.02 \text{ m}} = 3300 \text{ N}$$

component AD



$$\sum F_x = 0 \Rightarrow F_D = F_A = 3300 \text{ N}$$

$3300 \text{ N} > 2200 \text{ N}$ so the design outputs sufficient force to crack a nut.